Documentation for Active Hybrid Mooring (AHM) Rehearsal Simulation

August 18, 2025

Aero-Hydro-RTHS

$$\mathbf{M} \cdot \ddot{\mathbf{u}} + \mathbf{C} \cdot \dot{\mathbf{u}} + \mathbf{P_r}(\mathbf{u}, \dot{\mathbf{u}}) = \mathbf{P}(t)$$

Dynamic Loading:
- Seismic
- Wind
- Blast/Impact
- Wave
- Traffic

Static Loading:
- Gravity

Turbine Control:

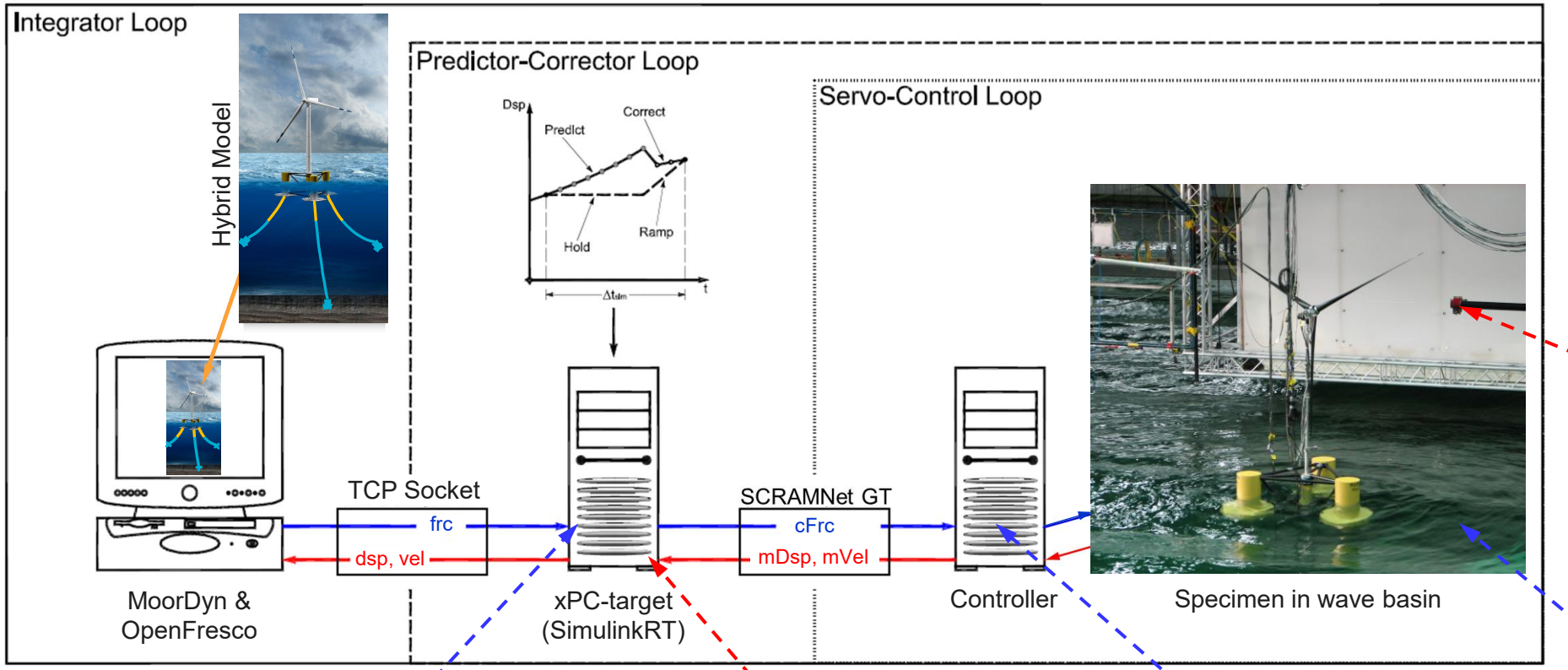


partitioning points

physical model of floating platform, upper portion of mooring lines, and hydrodynamic loads

analytical model of lower portion of mooring lines and anchors (optionally: structural tower, turbine, rotor, and aerodynamic loads)

Hybrid model and three-loop test architecture: **real test**



motion (dsp & vel) measurements

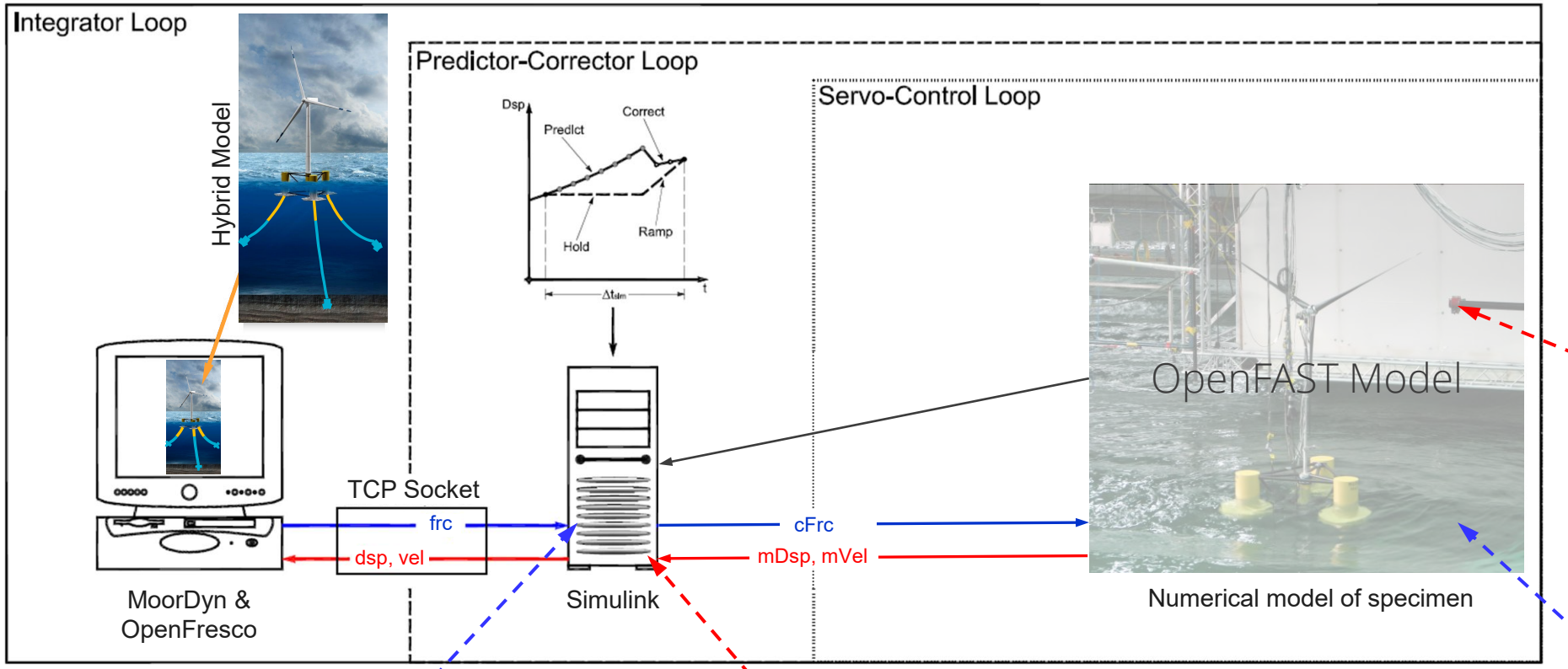
ATS delay compensator

filtering & noise reduction

TVC or other adv. ctrl. & force balancing

force actuation at the base of the wave basin

Hybrid model and three-loop test architecture: **rehearsal simulation**



motion (dsp & vel) calculated by numerical model

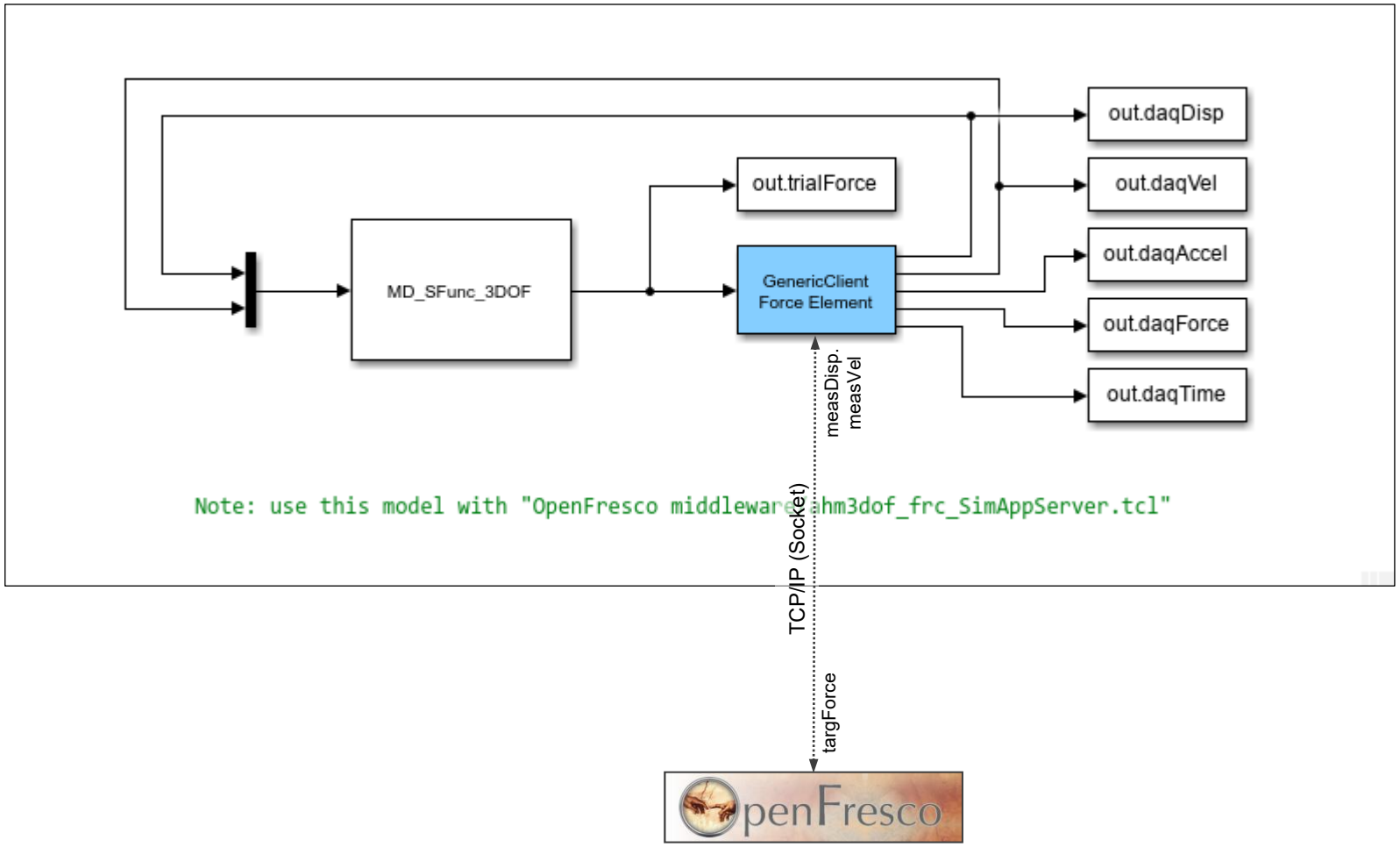
ATS delay compensator

filtering & noise reduction

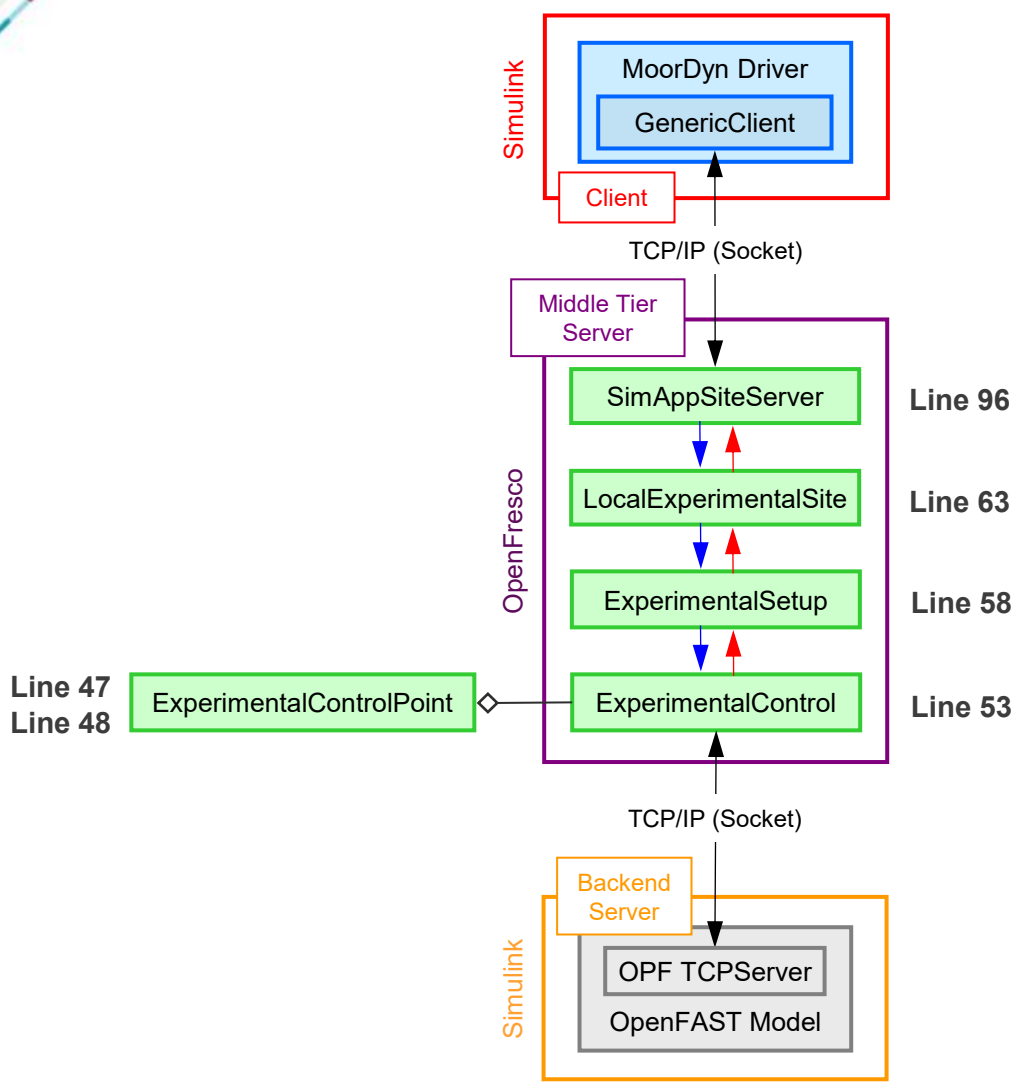
force actuation of the numerical model at the base of the wave basin

MoorDyn driver (integrator loop)

MoorDyn driver/MD_sfunc_test_3dof_opfFrc.slx



OpenFresco middleware (integrator loop, using TCL)



OpenFresco middleware/ahm3dof_frc_SimAppServer.tcl

```
# Define control points
# ---------------------
# expControlPoint $tag <-node $nodeTag> $dof $rspType <-fact $f> <-lim $l $u> <-relTrial> <-relCtrl> <-relDaq> ...
expControlPoint 1  1 force -relTrial 2 force -relTrial 3 force -relTrial 4 force -relTrial 5 force -relTrial 6 force -relTrial 7 force -relTrial
expControlPoint 2  1 force 2 force 3 force 4 force 5 force 6 force 7 force 8 force 9 force 1 disp  2 disp 3 disp 4 disp 5 disp 6 disp 7 disp 8

# Define experimental control
# ---------------------------
# expControl SimSimulink $tag ipAddr $ipPort -trialCP $cpTags -outCP $cpTags
expControl SimSimulink 1 "127.0.0.1" 22222 -trialCP 1 -outCP 2

# Define experimental setup
# -------------------------
# expSetup NoTransformation $tag <-control $ctrlTag> -dof $DOFs -sizeTrialOut $t $o <-trialDispFact $f> ...
expSetup NoTransformation 1 -control 1 -dof 1 2 3 4 5 6 7 8 9 -sizeTrialOut 9 9

# Define experimental site
# ------------------------
# expSite LocalSite $tag $setupTag
expSite LocalSite 1 1
# setSizeExpSite $siteTag -ctrlSizes (5 $size) -daqSizes (5 &size)
setSizeExpSite 1 -ctrlSizes 0 0 0 9 1 -daqSizes 9 9 9 9 1

# ------------------------------
# End of model generation
# ------------------------------


# ------------------------------
# Start of recorder generation
# ------------------------------
expRecorder Site -file "Output/Site_trialFrc.out" -time -site 1 trialForce
expRecorder Site -file "Output/Site_outDsp.out" -time -site 1 outDisp
expRecorder Site -file "Output/Site_outVel.out" -time -site 1 outVel
expRecorder Site -file "Output/Site_outFrc.out" -time -site 1 outForce

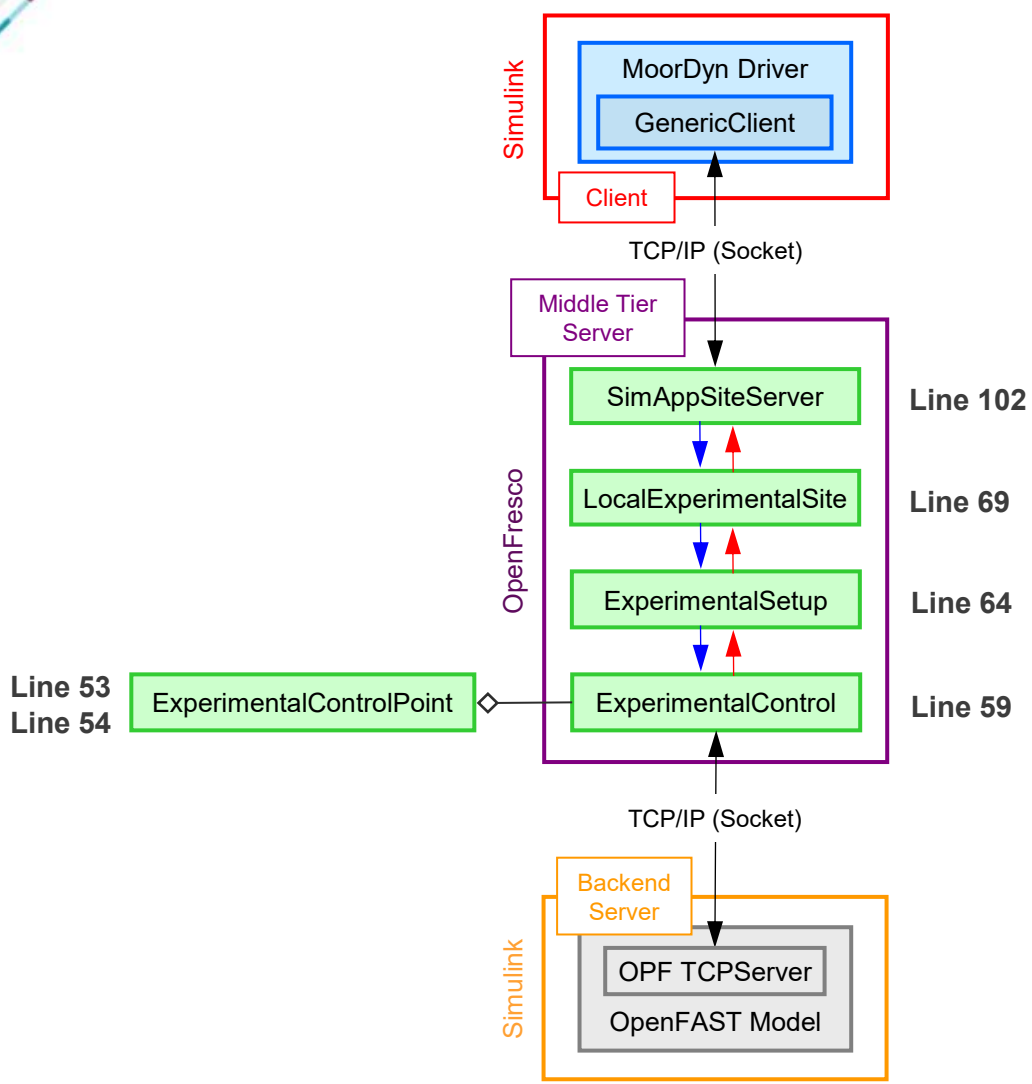
expRecorder Setup -file "Output/Setup_trialFrc.out" -time -setup 1 trialForce
expRecorder Setup -file "Output/Setup_outDsp.out" -time -setup 1 outDisp
expRecorder Setup -file "Output/Setup_outVel.out" -time -setup 1 outVel
expRecorder Setup -file "Output/Setup_outFrc.out" -time -setup 1 outForce

expRecorder Control -file "Output/Control_ctrlSig.out" -time -control 1 ctrlSignal
expRecorder Control -file "Output/Control_daqSig.out" -time -control 1 daqSignal
# ---------------------------------
# End of recorder generation
# ---------------------------------


# ------------------------------
# Start the server process
# ------------------------------
# startSimAppSiteServer $siteTag $port <-ssl> <-udp>
startSimAppSiteServer 1 8090;

wipeExp
exit
```

OpenFresco middleware (integrator loop, using Python)



OpenFresco middleware/ahm3dof_frc_SimAppServer.py

```
# Define control points
# --------------------
# expControlPoint(tag, <"-node", nodeTag,> dof, rspType, <"-fact", f,> <"-lim", l, u,> <"-relTrial",> <"-relCtrl",> <"-relDaq",> ...)
expControlPoint(1, "1", "force", "-relTrial", "2", "force", "-relTrial", "3", "force", "-relTrial", "4", "force", "-relTrial", "5", "for
expControlPoint(2, "1", "force", "2", "force", "3", "force", "4", "force", "5", "force", "6", "force", "7", "force", "8", "force", "9",

# Define experimental control
# ---------------------------
# expControl("SimSimulink", tag, "ipAddr", ipPort, "-trialCP", cpTags, "-outCP", cpTags)
expControl("SimSimulink", 1, "127.0.0.1", 22222, "-trialCP", 1, "-outCP", 2)

# Define experimental setup
# -------------------------
# expSetup("NoTransformation", tag, <"-control", ctrlTag,> "-dof", DOFs, "-sizeTrialOut", t, o, <"-trialDispFact", $f,> ...)
expSetup("NoTransformation", 1, "-control", 1, "-dof", 1, 2, 3, 4, 5, 6, 7, 8, 9, "-sizeTrialOut", 9, 9)

# Define experimental site
# ------------------------
# expSite("LocalSite", tag, setupTag)
expSite("LocalSite", 1, 1)
# setSizeExpSite(siteTag, "-ctrlSizes", (5 size), "-daqSizes", (5 size))
setSizeExpSite(1, "-ctrlSizes", 0, 0, 0, 9, 1, "-daqSizes", 9, 9, 9, 9, 1)

# ----------------------------
# End of model generation
# ----------------------------


# ----------------------------
# Start of recorder generation
# ----------------------------
expRecorder("Site", "-file", "Output/Site_trialFrc.out", "-time", "-site", 1, "trialForce")
expRecorder("Site", "-file", "Output/Site_outDsp.out", "-time", "-site", 1, "outDisp")
expRecorder("Site", "-file", "Output/Site_outVel.out", "-time", "-site", 1, "outVel")
expRecorder("Site", "-file", "Output/Site_outFrc.out", "-time", "-site", 1, "outForce")

expRecorder("Setup", "-file", "Output/Setup_trialFrc.out", "-time", "-setup", 1, "trialForce")
expRecorder("Setup", "-file", "Output/Setup_outDsp.out", "-time", "-setup", 1, "outDisp")
expRecorder("Setup", "-file", "Output/Setup_outVel.out", "-time", "-setup", 1, "outVel")
expRecorder("Setup", "-file", "Output/Setup_outFrc.out", "-time", "-setup", 1, "outForce")

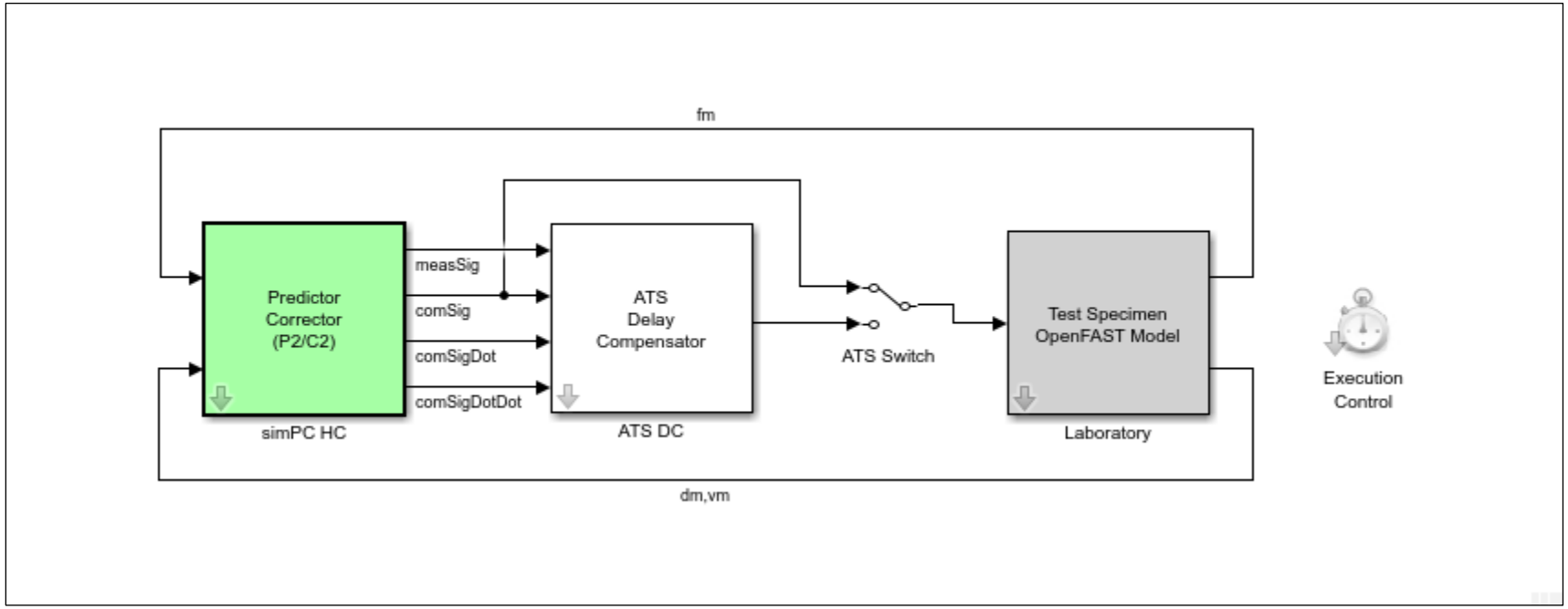
expRecorder("Control", "-file", "Output/Control_ctrlSig.out", "-time", "-control", 1, "ctrlSignal")
expRecorder("Control", "-file", "Output/Control_daqSig.out", "-time", "-control", 1, "daqSignal")
# --------------------------------
# End of recorder generation
# --------------------------------


# ----------------------------
# Start the server process
# ----------------------------
# startSimAppSiteServer(siteTag, port, <"-ssl",> <"-udp">)
startSimAppSiteServer(1, 8090);  # use with experimental element in FEA

wipeExp()
exit()
```

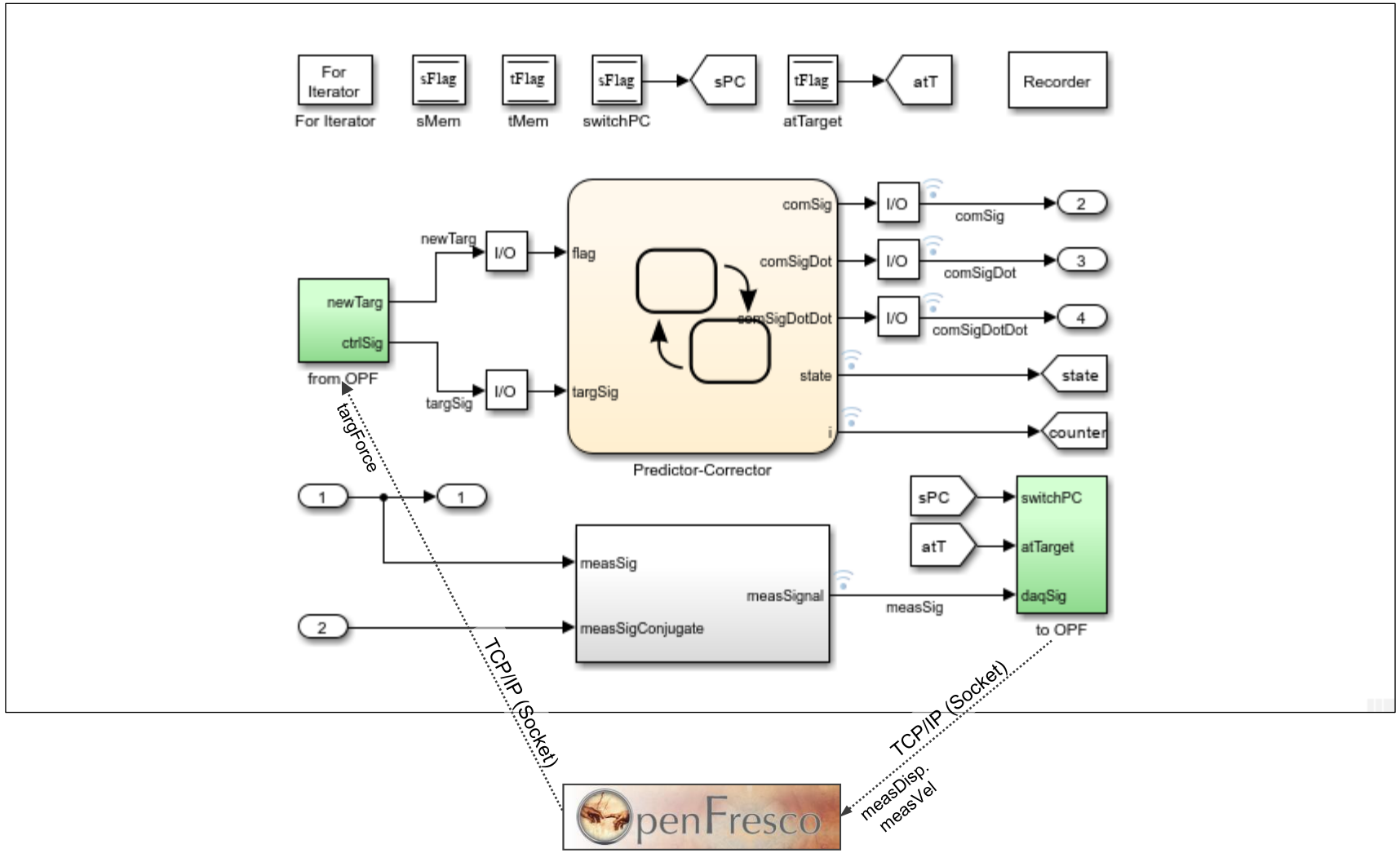
Laboratory simulation

Laboratory simulation/HybridControllerP2C2ahm.slx



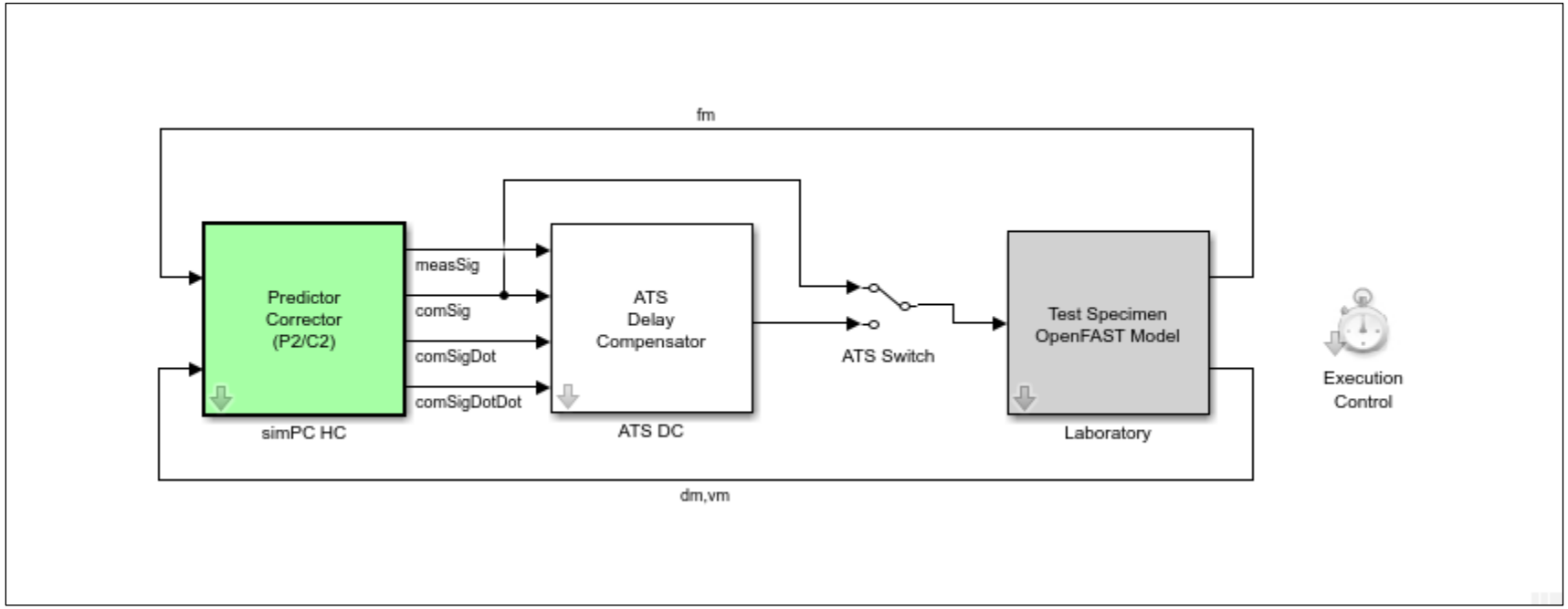
Laboratory simulation: Predictor-Corrector (P2/C2)

Laboratory simulation/HybridControllerP2C2ahm.slx -> simPC HC



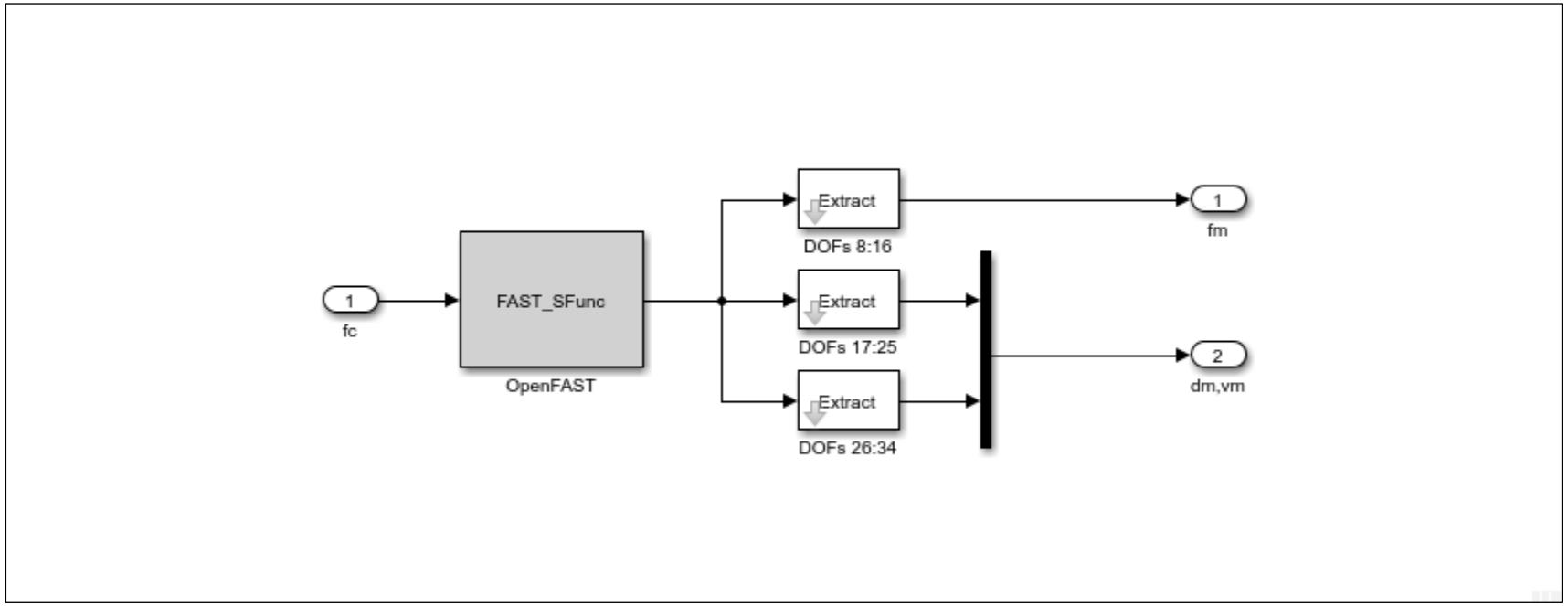
Laboratory simulation

Laboratory simulation/HybridControllerP2C2ahm.slx



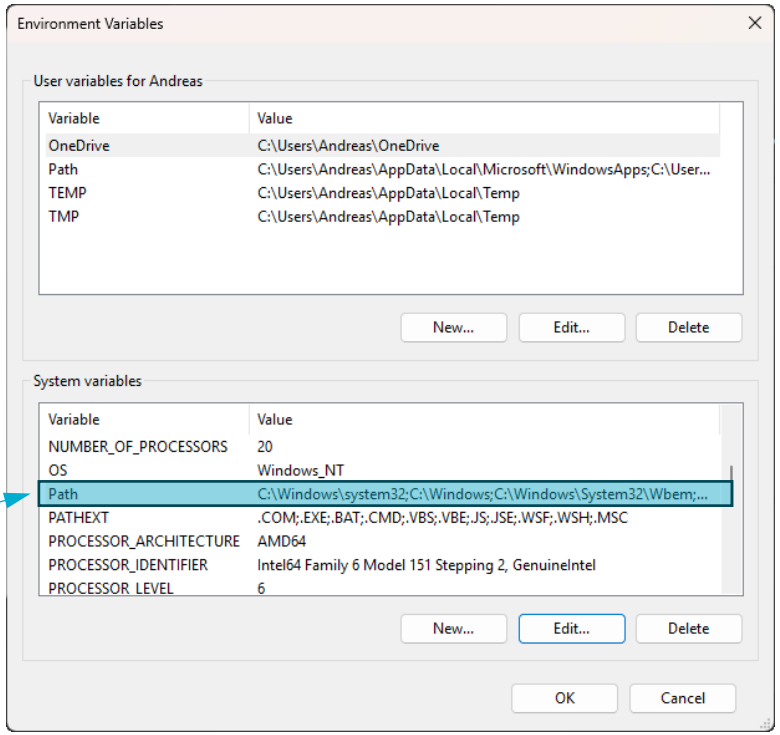
Laboratory simulation: OpenFAST model of specimen

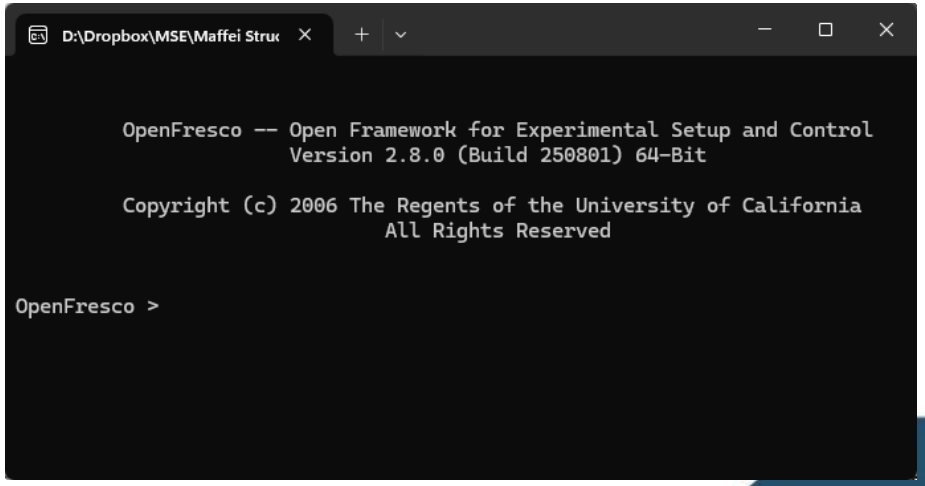
Laboratory simulation/HybridControllerP2C2ahm.slx -> Laboratory



Installing the software

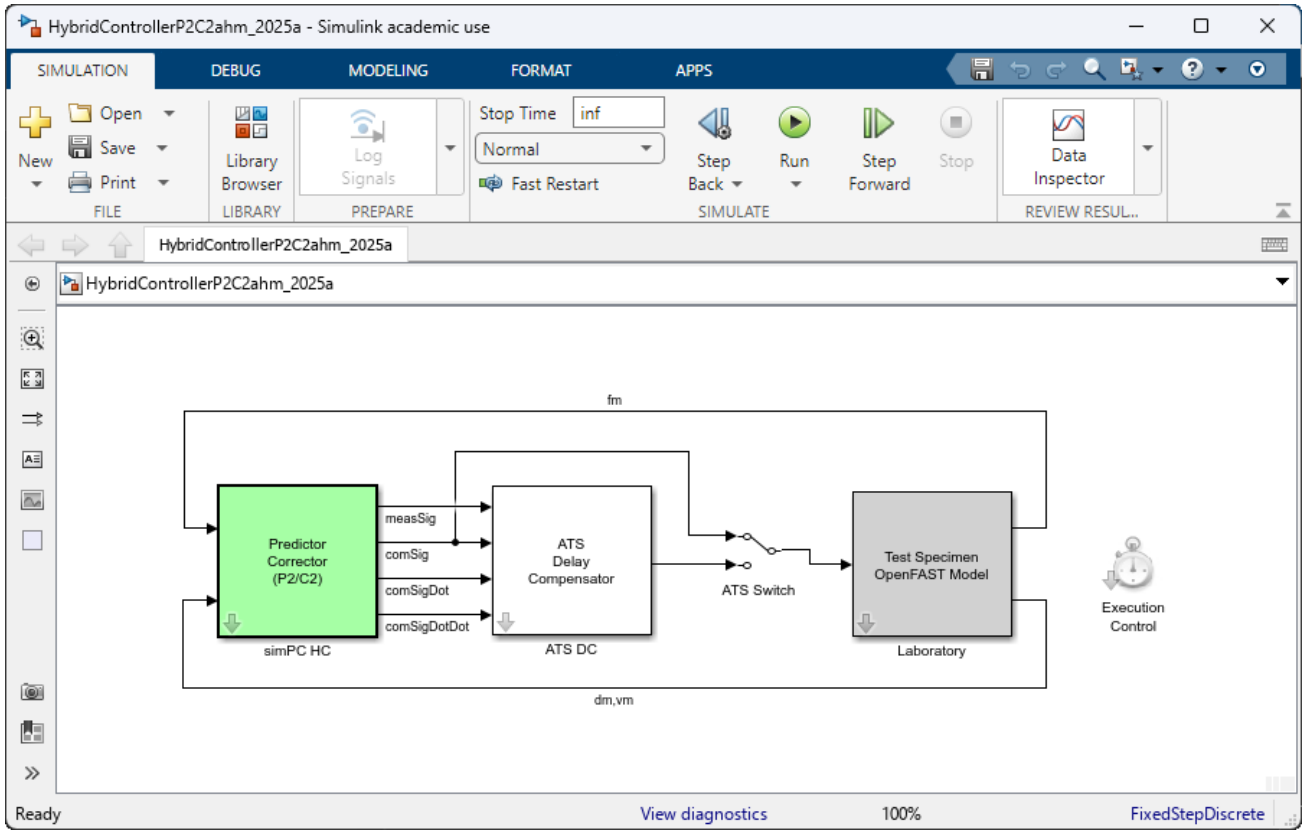
1. From the “OpenFresco middleware/Software” directory, install Tcl and OpenSSL by executing the installers.
   a) Install Tcl in the directory “C:\Program Files\Tcl”
   b) Install OpenSSL in the directory “C:\Program Files\OpenSSL”
2. Open Matlab and add the folder “OpenFresco middleware/Software/HybridSimToolbox” to the path.
3. Add the folder “OpenFresco middleware” to the system path variable. Go to Settings->System->About->Advanced System Settings->Environment Variables…->System variables->Path. Click Edit… and add the folder.
4. Test OpenFresco by double-clicking on “OpenFrescoTcl.exe” in the “OpenFresco middleware” directory. It should look like the figure on the right.





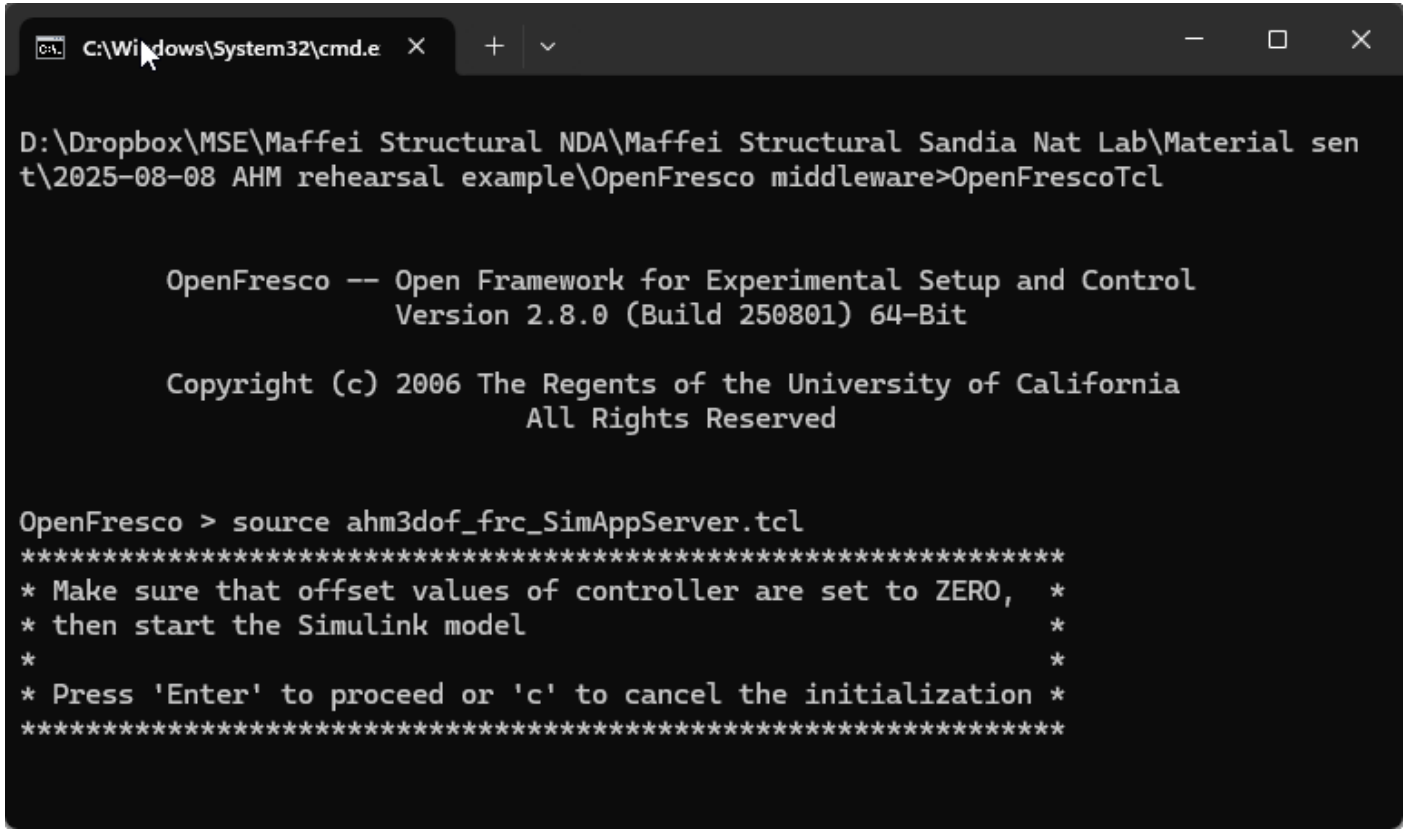
Running the simulation

1. Prepare the laboratory simulation (back-end server):
   a) Open 1st instance of Matlab and change the folder to “Laboratory simulation”.
   b) Open the Simulink model “HybridControllerP2C2ahm.slx”.



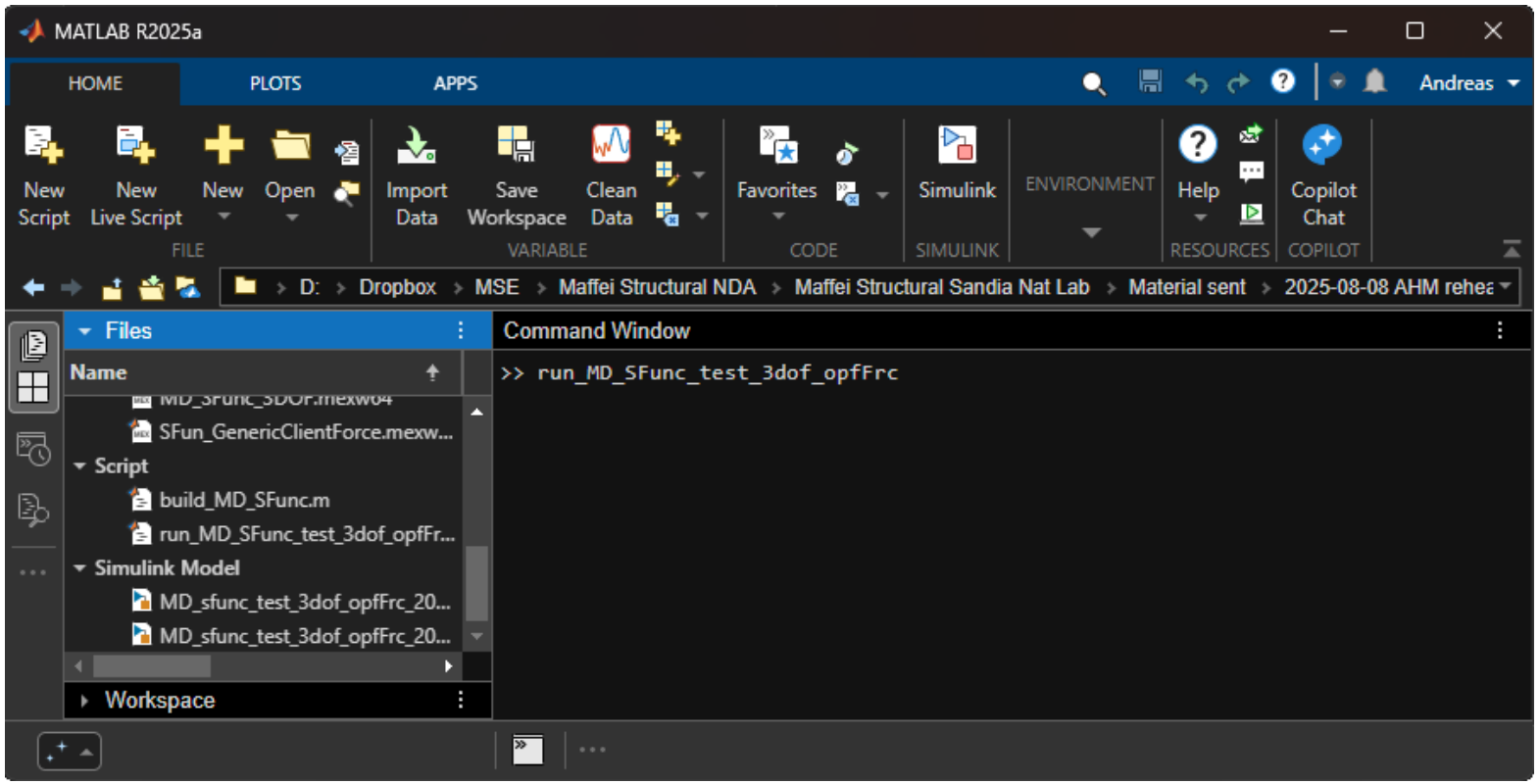
Running the simulation

2. Prepare the OpenFresco middleware (middle-tier server):
   a) Open a Command Prompt in the folder “OpenFresco middleware”.
   b) Start OpenFresco by typing “OpenFrescoTcl” and pressing Enter.
   c) Run the Tcl script by typing “source ahm3dof_frc_SimAppServer.tcl” and pressing Enter.



Running the simulation

3. Prepare the MoorDyn driver (client):
   a) Open 2nd instance of Matlab and change the folder to “MoorDyn driver”.
   b) Open the Matlab script “run_MD_SFunc_test_3dof_opfFrc.m” and check that Line 15 calls the correct MoorDyn Simulink model.
   c) Switch to the Matlab command prompt and type “run_MD_SFunc_test_3dof_opfFrc” but do NOT press Enter yet.



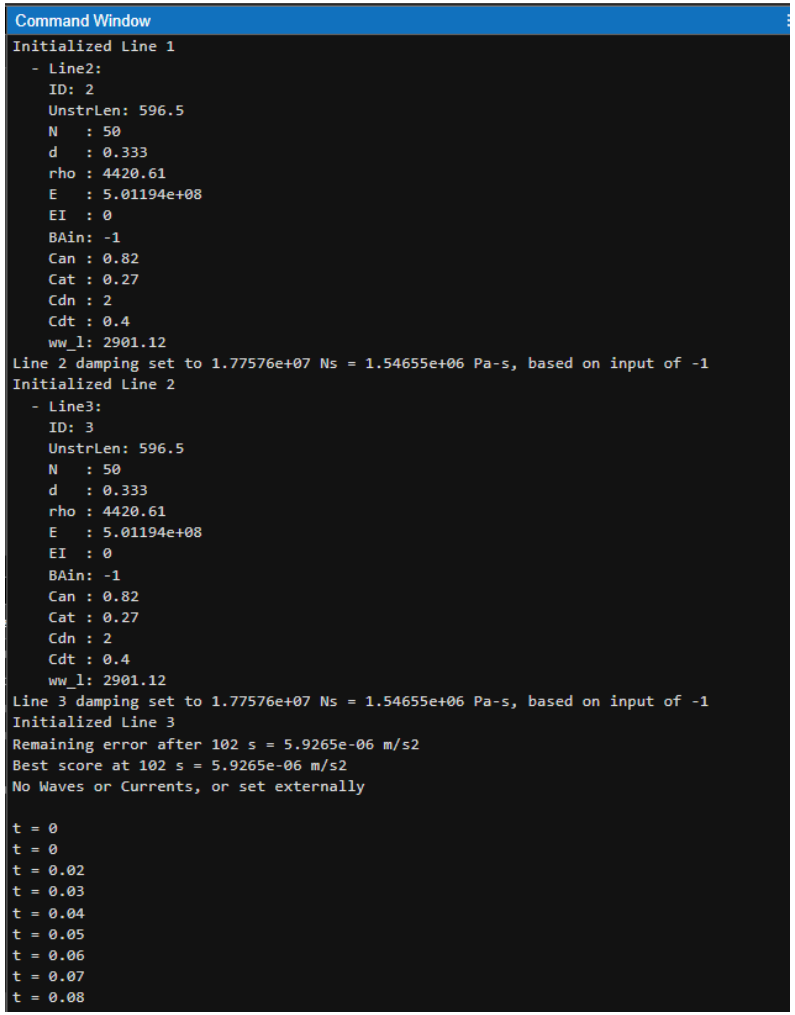
Running the simulation

4. Running the simulation:
   a) Switch to the “HybridControllerP2C2ahm.slx” model of the 1st Matlab instance and click “Run” to start the back-end server.
   b) Switch to the OpenFresco command prompt and press Enter. It will display the ExperimentalControl information and the initial values of the DAQ. If you are satisfied with the initial DAQ values, press Enter again to start the middle-tier server.
   c) Switch to the 2nd Matlab command prompt where you typed “run_MD_SFunc_test_3dof_opfFrc” and press Enter to start the client.



Running the simulation

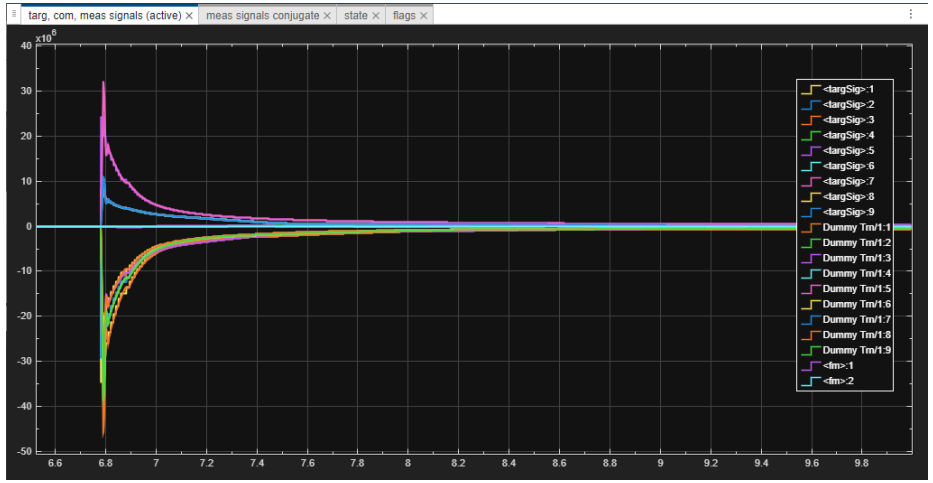
5. Here is what you should see while the simulation is running

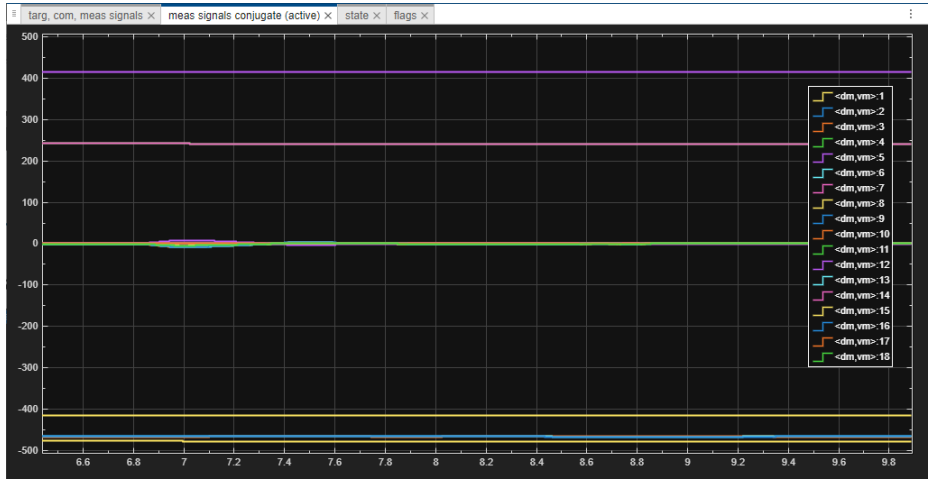


Client (MoorDyn driver)



Middle-tier server (OpenFresco)

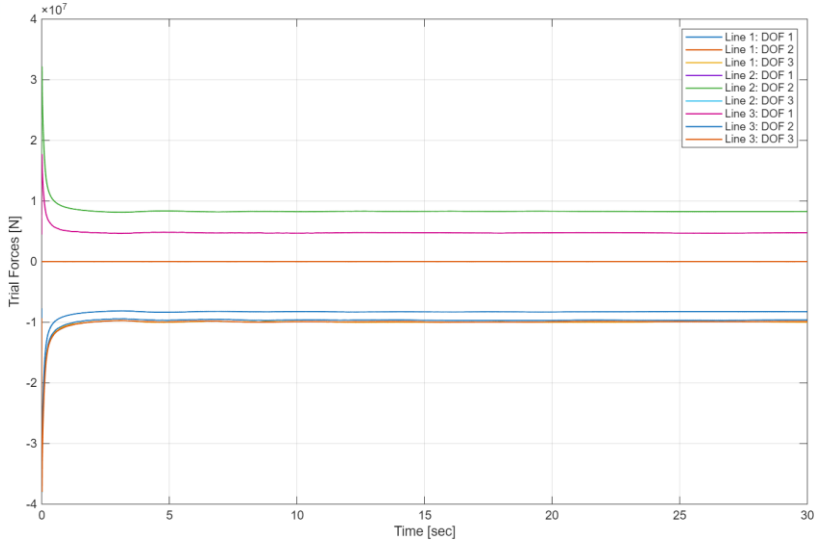


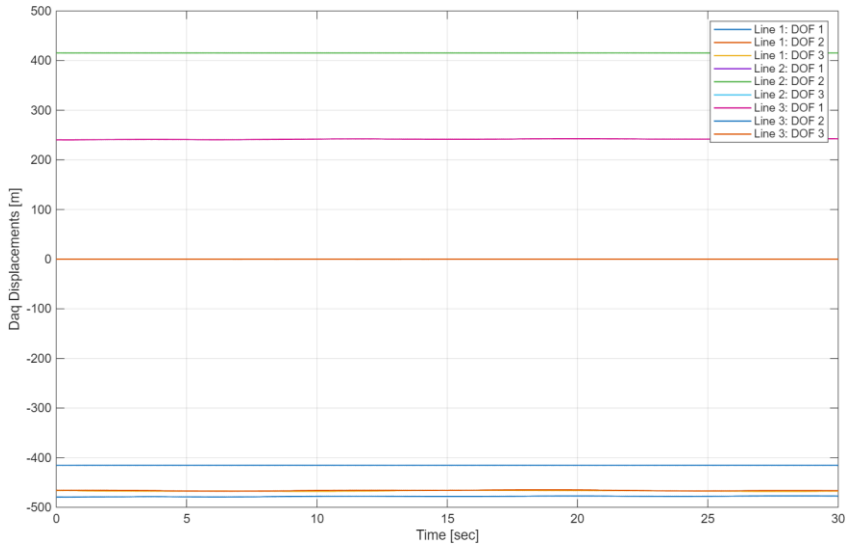


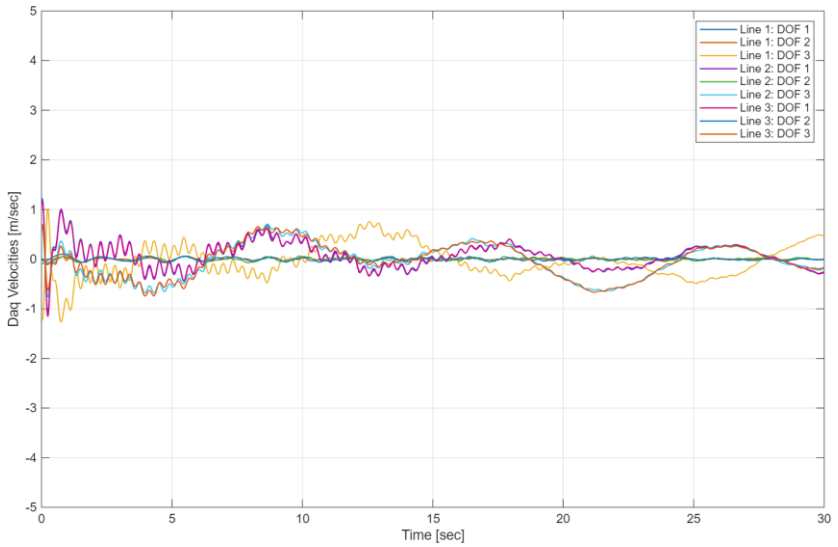
Back-end server (Lab simulation)

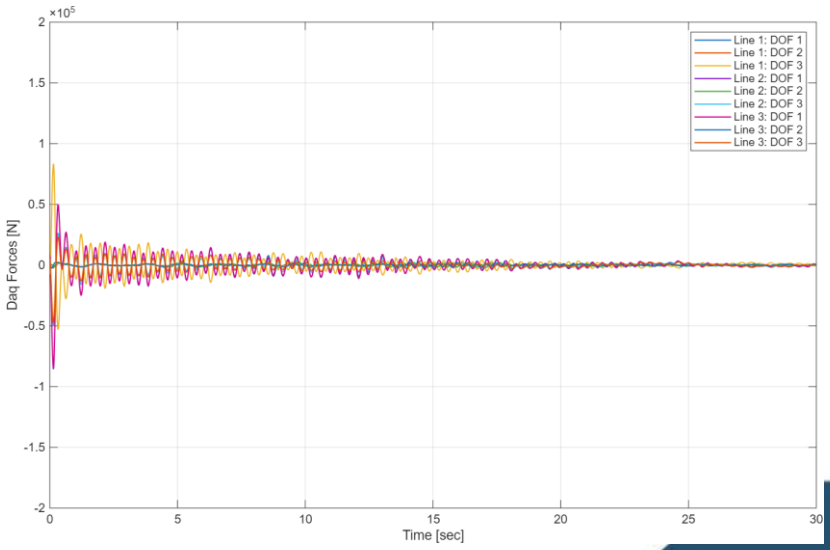
Running the simulation

6. Output from the MoorDyn driver









Running the simulation

7. Output from the OpenFresco middleware

Control_ctrlSig.out

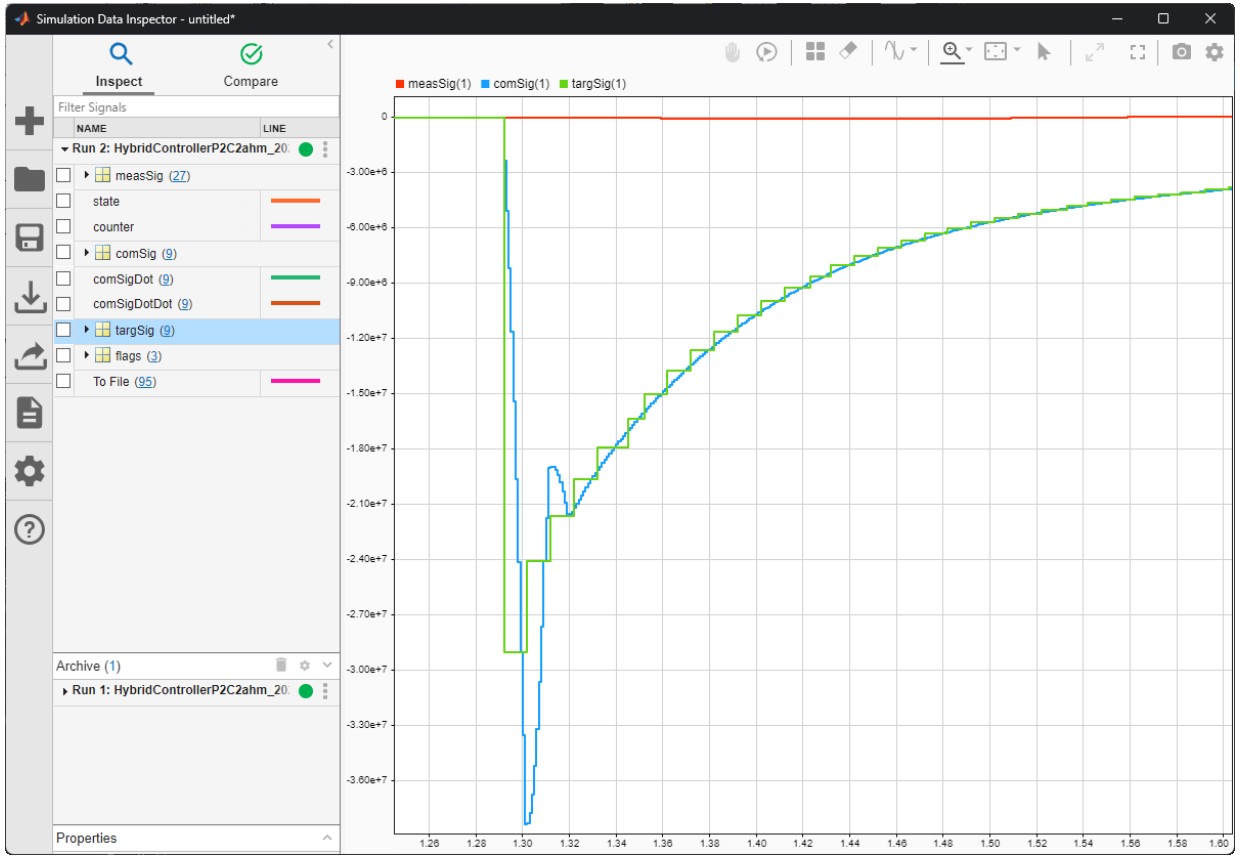
```
0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0
0.01 -3.46026e+07 -15293.4 -2.35553e+07 8.18093e+06 2.41779e+07 -2.90858e+07 8.25356e+06 -2.4027e+07 -2.91346e+07 0.01
0.02 -2.87986e+07 -13846.9 -1.9283e+07 6.57117e+06 1.99694e+07 -2.40155e+07 6.63336e+06 -1.98416e+07 -2.40571e+07 0.02
0.03 -2.58416e+07 -13168.1 -1.73017e+07 5.86264e+06 1.78436e+07 -2.14512e+07 5.91768e+06 -1.77294e+07 -2.14863e+07 0.03
0.04 -2.34318e+07 -12869.3 -1.58166e+07 5.38114e+06 1.61623e+07 -1.9421e+07 5.43024e+06 -1.60616e+07 -1.94521e+07 0.04
0.05 -2.1322e+07 -13156 -1.45863e+07 5.01639e+06 1.47161e+07 -1.76726e+07 5.05952e+06 -1.46263e+07 -1.76985e+07 0.05
0.06 -1.94377e+07 -14248.7 -1.352e+07 4.7222e+06 1.34397e+07 -1.61269e+07 4.76e+06 -1.3359e+07 -1.61484e+07 0.06
0.07 -1.77422e+07 -16401.7 -1.25709e+07 4.47276e+06 1.23012e+07 -1.47445e+07 4.50601e+06 -1.22276e+07 -1.4763e+07 0.07
0.08 -1.6211e+07 -19909.1 -1.17117e+07 4.25272e+06 1.12807e+07 -1.35008e+07 4.28214e+06 -1.12118e+07 -1.35181e+07 0.08
0.09 -1.48263e+07 -25094.2 -1.09261e+07 4.05291e+06 1.0364e+07 -1.23777e+07 4.07924e+06 -1.02975e+07 -1.23961e+07 0.09
0.1 -1.35738e+07 -32266.1 -1.02042e+07 3.86833e+06 9.54056e+06 -1.13617e+07 3.89224e+06 -9.47395e+06 -1.13838e+07 0.1
0.11 -1.23167e+07 -61218.8 -9.52884e+06 3.73729e+06 8.77457e+06 -1.03462e+07 3.76001e+06 -8.6909e+06 -1.03903e+07 0.11
0.12 -1.12933e+07 -75909.1 -8.92971e+06 3.58732e+06 8.11853e+06 -9.51145e+06 3.60864e+06 -8.0281e+06 -9.56773e+06 0.12
0.13 -1.03736e+07 -91950.3 -8.38214e+06 3.44745e+06 7.53273e+06 -8.75808e+06 3.46752e+06 -7.43426e+06 -8.82822e+06 0.13
0.14 -9.55035e+06 -108652 -7.88467e+06 3.3175e+06 7.01137e+06 -8.08061e+06 3.33633e+06 -6.90421e+06 -8.16555e+06 0.14
0.15 -8.81631e+06 -125191 -7.43538e+06 3.19713e+06 6.54856e+06 -7.4738e+06 3.21458e+06 -6.4328e+06 -7.5736e+06 0.15
0.16 -8.16426e+06 -140722 -7.03172e+06 3.08565e+06 6.13832e+06 -6.93246e+06 3.10154e+06 -6.0148e+06 -7.04627e+06 0.16
0.17 -7.58714e+06 -154478 -6.67056e+06 2.98209e+06 5.77496e+06 -6.45151e+06 2.99614e+06 -5.64496e+06 -6.57759e+06 0.17
0.18 -7.07805e+06 -165849 -6.34806e+06 2.88524e+06 5.45308e+06 -6.02589e+06 2.89729e+06 -5.31844e+06 -6.16212e+06 0.18
0.19 -6.63047e+06 -174434 -6.06017e+06 2.79413e+06 5.16766e+06 -5.65058e+06 2.80395e+06 -5.03026e+06 -5.79413e+06 0.19
0.2 -6.23796e+06 -180107 -5.80272e+06 2.70823e+06 4.91407e+06 -5.32014e+06 2.71575e+06 -4.77598e+06 -5.46823e+06 0.2
0.21 -5.89416e+06 -182976 -5.57235e+06 2.62743e+06 4.68803e+06 -5.02927e+06 2.63262e+06 -4.55112e+06 -5.17906e+06 0.21
0.22 -5.59231e+06 -183275 -5.36592e+06 2.55167e+06 4.48538e+06 -4.77267e+06 2.55459e+06 -4.35146e+06 -4.92157e+06 0.22
0.23 -5.32611e+06 -181329 -5.18053e+06 2.48097e+06 4.30252e+06 -4.54534e+06 2.48168e+06 -4.17303e+06 -4.69102e+06 0.23
0.24 -5.0895e+06 -177506 -5.01351e+06 2.41533e+06 4.13635e+06 -4.34291e+06 2.41394e+06 -4.01255e+06 -4.48337e+06 0.24
0.25 -4.87734e+06 -172183 -4.86255e+06 2.35458e+06 3.98413e+06 -4.16134e+06 2.35125e+06 -3.86695e+06 -4.29497e+06 0.25
0.26 -4.68503e+06 -165712 -4.72531e+06 2.29852e+06 3.84353e+06 -3.99713e+06 2.29337e+06 -3.7338e+06 -4.12292e+06 0.26
0.27 -4.50879e+06 -158406 -4.59988e+06 2.24687e+06 3.71288e+06 -3.84772e+06 2.24001e+06 -3.61099e+06 -3.96466e+06 0.27
0.28 -4.34577e+06 -150526 -4.48465e+06 2.1992e+06 3.59059e+06 -3.7107e+06 2.19078e+06 -3.49698e+06 -3.81845e+06 0.28
0.29 -4.19374e+06 -142281 -4.37811e+06 2.15507e+06 3.47578e+06 -3.58456e+06 2.14521e+06 -3.39061e+06 -3.68277e+06 0.29
0.3 -4.05119e+06 -133831 -4.27906e+06 2.11399e+06 3.36765e+06 -3.46789e+06 2.10274e+06 -3.29092e+06 -3.5566e+06 0.3
0.31 -3.91717e+06 -125291 -4.18653e+06 2.07532e+06 3.26548e+06 -3.35962e+06 2.06278e+06 -3.19724e+06 -3.43907e+06 0.31
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0.33 -3.67247e+06 -108234 -4.0173e+06 2.003e+06 3.07778e+06 -3.16576e+06 1.98794e+06 -3.02588e+06 -3.22725e+06 0.33
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0.54 -2.45749e+06 26927.4 -2.62011e+06 1.21388e+06 1.94315e+06 -2.13309e+06 1.17318e+06 -1.98255e+06 -2.10165e+06 0.54
0.55 -2.43682e+06 28689 -2.55901e+06 1.17696e+06 1.91162e+06 -2.10789e+06 1.13555e+06 -1.95049e+06 -2.07666e+06 0.55
0.56 -2.41795e+06 30048.2 -2.49794e+06 1.13996e+06 1.88123e+06 -2.0838e+06 1.09812e+06 -1.91924e+06 -2.05337e+06 0.56
0.57 -2.40055e+06 31873.9 -2.43734e+06 1.10271e+06 1.85159e+06 -2.06113e+06 1.06079e+06 -1.88875e+06 -2.03165e+06 0.57
0.58 -2.38439e+06 34126.2 -2.37727e+06 1.06512e+06 1.82227e+06 -2.03956e+06 1.02358e+06 -1.85896e+06 -2.01134e+06 0.58
0.59 -2.3692e+06 36665.6 -2.31758e+06 1.02727e+06 1.79321e+06 -2.01892e+06 986435 -1.82965e+06 -1.99204e+06 0.59
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Control_daqSig.out

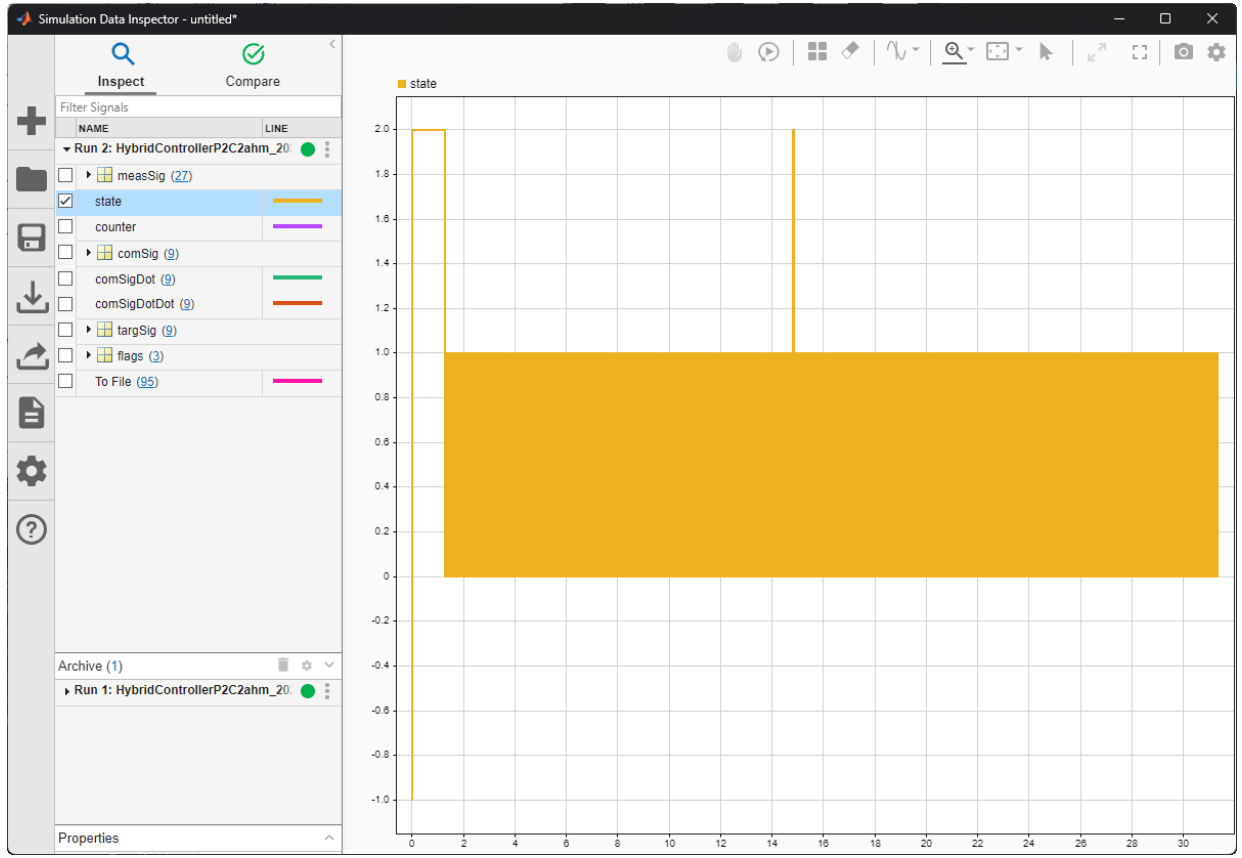
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0 -1246.46 879.557 1114.87 -1207.68 663.826 794.244 -1070.26 1352.47 -977.247 -476.875 0.00490294 -468.439 242.966 415.541
0.01 -1678.62 866.691 457.248 -767.95 1092.85 291.247 -652.94 844.764 -1437.68 -476.88 0.00519067 -468.433 242.961 415.542
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0.03 -39959.3 941.356 29781.8 -31558.1 4979.23 -21340.9 -31216.1 -2612.17 -23451.8 -476.89 0.00582114 -468.422 242.951 415.5
0.04 -74586.4 1005.79 58409.9 -61308.3 7225.04 -39985.6 -60750.9 -4388.42 -42426.4 -476.896 0.00616541 -468.416 242.945 415.
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```

Running the simulation

8. Output from the laboratory simulation



targ vs. comm vs. meas signals



state of predictor-corrector
(0 = correct, 1 = predict, 2 = slow-down)

Compiling MoorDyn with VS2022

1. Fork MoorDyn from the following GitHub repository: https://github.com/FloatingArrayDesign/MoorDyn
2. Follow the instructions here (https://moordyn.readthedocs.io/en/latest/compiling.html#compile-using-cmake) to compile MoorDyn using CMake.
3. Move the “moordyn.dll” and “moordyn.lib files from the build/source/Release directory to the “Laboratory simulation” folder.
4. Use the file build_MD_SFunc.m in the folder “MoorDyn driver” to build the S-Function “MD_Sfunc_3DOF”. Ensure that you adjust Lines 22 and 23 according to where you built MoorDyn.

Compiling OpenFAST with VS2022

1. Fork the modified OpenFAST repository from Michael Devin (make sure to use the ahm branch): https://github.com/michaelcdevin/openfast/tree/ahm
2. Open the vs-build/FAST/FAST.sln solution
3. In the Solution Explorer, right-click the top-level solution Solution 'FAST' (5 of 5 projects) and click Retarget solution, and hit OK on the new window with the defaults
4. In the OpenFAST-Simulink project properties, change the Linker/Input/Additional Dependencies to the path of your “libmex.lib” library for your version of Matlab on your computer. The path is likely something like C:\Program Files\MATLAB\[matlab version]\extern\lib\win64\microsoft\libmex.lib
5. Change the build to Release_Matlab and build the solution. The compiled OpenFAST shared library should be under build/bin/OpenFAST-Simulink_x64.dll
6. Open the glue-codes/simulink/src/create_FAST_SFunc.m file in Matlab and run it from its directory. The compiled S-Function should be under build/bin/FAST_SFunc.mexw64.
7. Move the “OpenFAST-Simulink_x64.dll”, “OpenFAST-Simulink_x64.lib”, and “FAST_SFunc.mexw64” files from the build/bin/ directory to the “Laboratory simulation” folder.

MD_sfunc_test_3dof_opfFrc - Simulink academic use

SIMULATION DEBUG MODELING FORMAT APPS

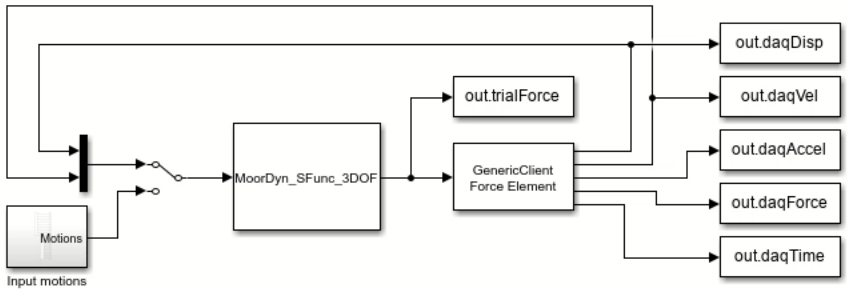
New Open Save Print | Library Browser | Log Signals | Stop Time TMax | Normal | Fast Restart | Step Back | Run | Step Forward | Stop | Data Inspector

FILE LIBRARY PREPARE SIMULATE REVIEW RESUL...

Model Browser

MD_sfunc_test_3dof_opfFrc

MD_sfunc_test_3dof_opfFrc



Note: use this model with "OpenFresco/ahm3dof_frc_SimAppServer.tcl"

Property Inspector

Ready 100% FixedStepDiscrete

C:\Windows\System32\cmd.e

```
C:\Users\Andreas\Documents\ActiveHybridMooring\OpenFresco>openfrescotcl

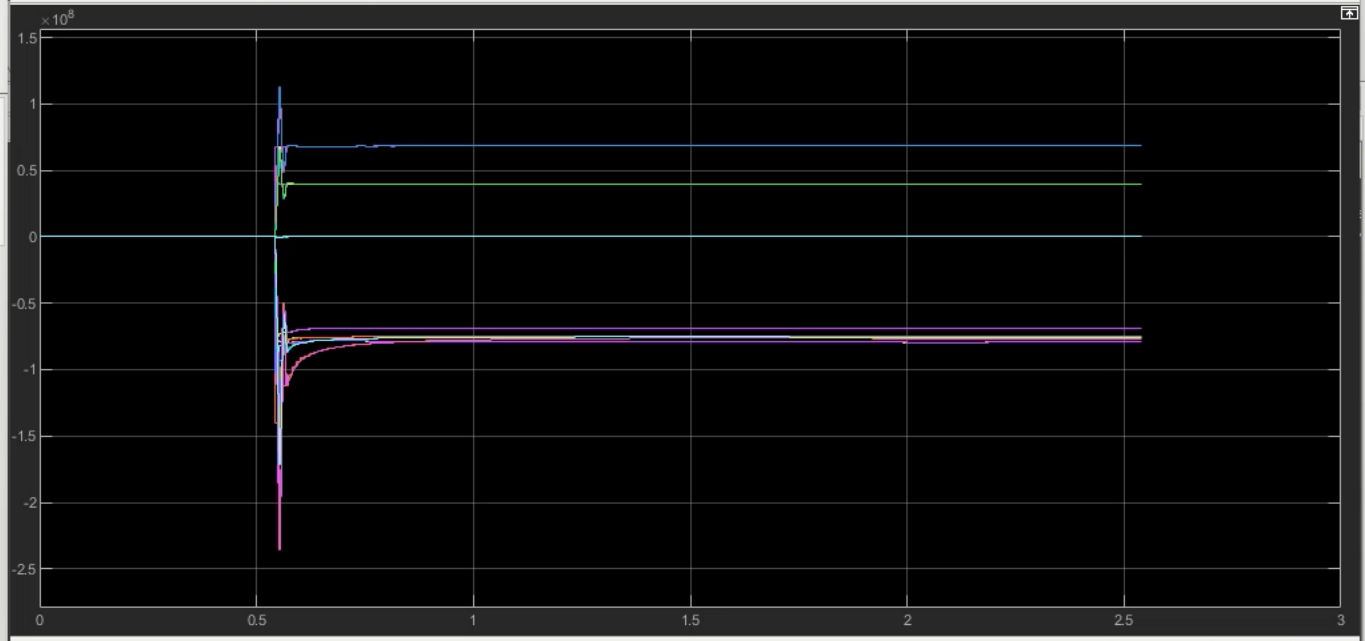
        OpenFresco -- Open Framework for Experimental Setup and Control
                      Version 2.8.0 (Build 240801) 64-Bit

        Copyright (c) 2006 The Regents of the University of California
                           All Rights Reserved

OpenFresco > source ahm3dof_frc_SimAppServer.tcl
```

targ, com, meas signals

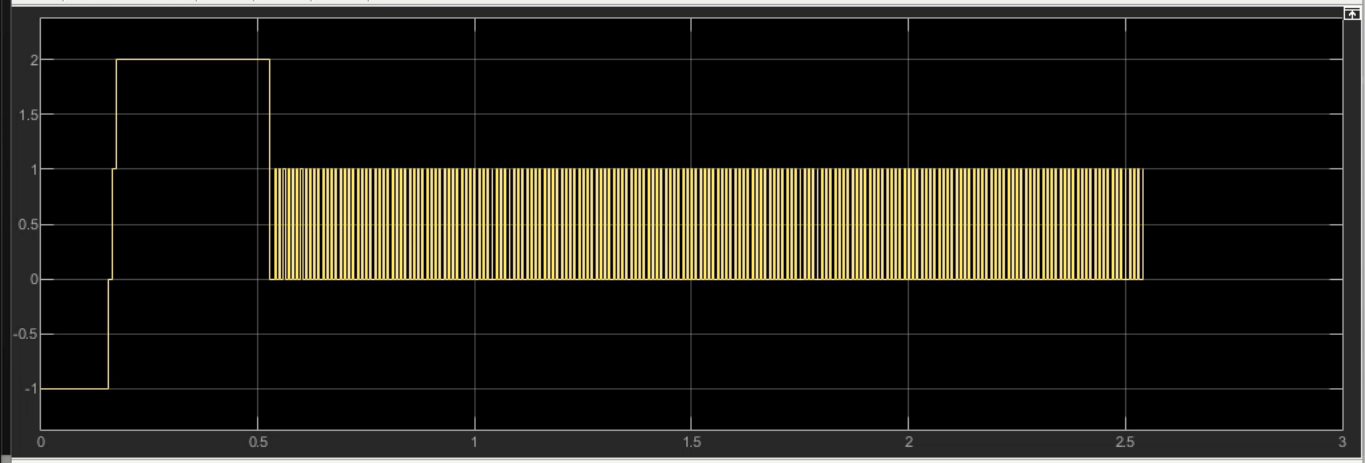
File Tools View Simulation Help



Ready Sample based Offset=0 T=2.540

state

File Tools View Simulation Help



Ready Sample based Offset=0 T=2.540

23:56
3/25/2025